



MAX-PLANCK-GESELLSCHAFT

Chemical data as stochastic processes

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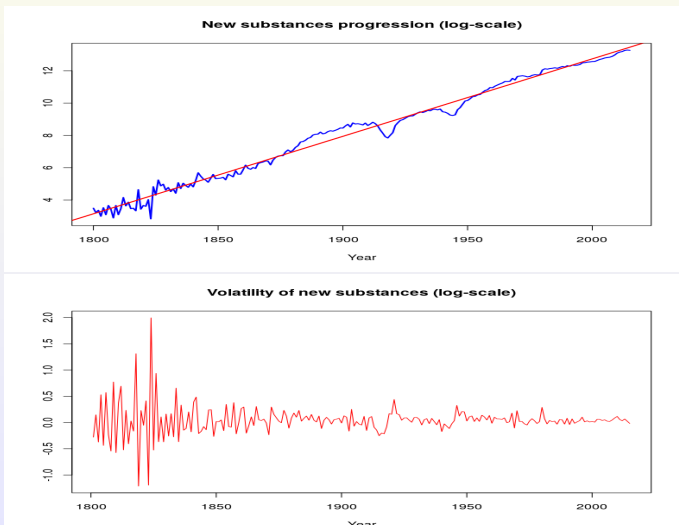
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Contents

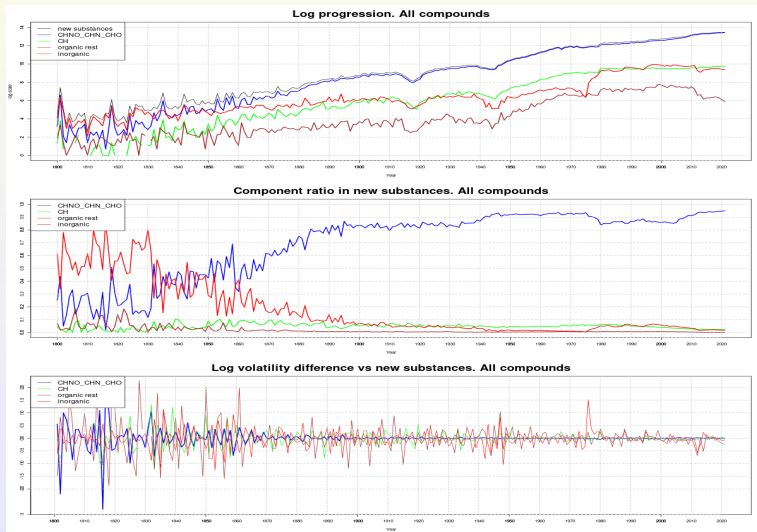
- The number of chemical formulae found every year is exponentially growing
- 95 percent of the new substances found every year belong to the **conservative area**, i.e. $|C| \leq 60$, $|H| \leq 120$, $|N| < 10$, $|O| < 20$.
- The volatility reduction is a result of capacity limit in the conservative area and the limit in geometric combination possibilities in each chemical formula. Upper bound 250.
- New trend starts from 1970 with remarkable growth in the **complexity area**, i.e. $|C| > 60$ in hydro-carbon (CH) that might act as the driving force for the exponential growth to sustain.
- Evolution of the number of new substances of the same formula shows similarity, which accounts for the long-range dependence.

Evolution of new substances: Exponential growth with reduction of variance

$$\text{Volatility } r_t := \log\left(\frac{N_{t+1}}{N_t}\right).$$



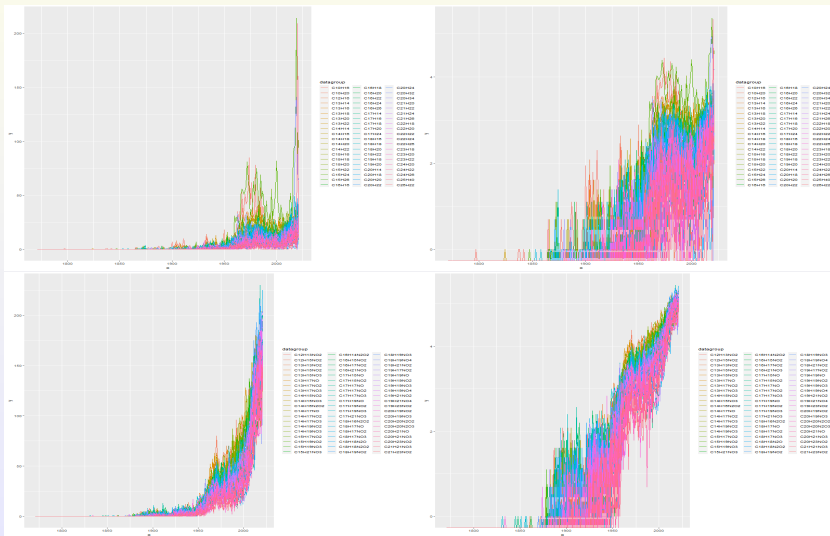
A zoom in



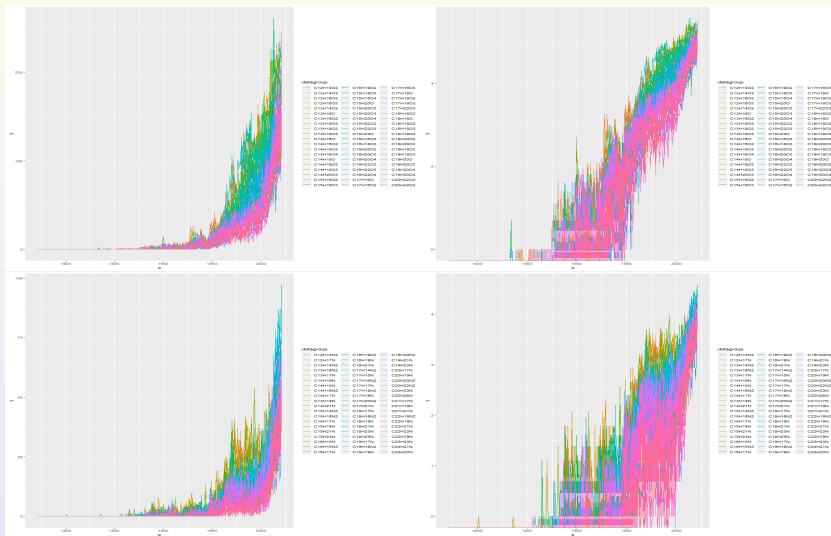
Data space

- Ω is the set of all chemical formulae, e.g. $C_xH_yN_zO_w$, $C_xH_yO_w$, $C_xH_yN_z$, C_xH_y .
- At time t , $\Omega(t) \subset \Omega$ is the set of every formula ω of new substances found, together with its multiplicity $y_i(t)$. $D(t)$ is the cardinality of $\Omega(t)$, and $V(t) = \sum y_i(t)$ is total of new substances found at time t .
- Formally, one can construct an abstract probability space $(\Omega, \mathcal{F}, \mathbb{P})$. But in practice it is important to check the stationarity assumption of the probability measure.
- One would like to analyze the dynamics of y_i over time as time series and their mutual relations to get conclusions on the dynamics of V_t .
- On the other hand, studying the formula's coordinates and their dynamics also reveals rich information on the population.

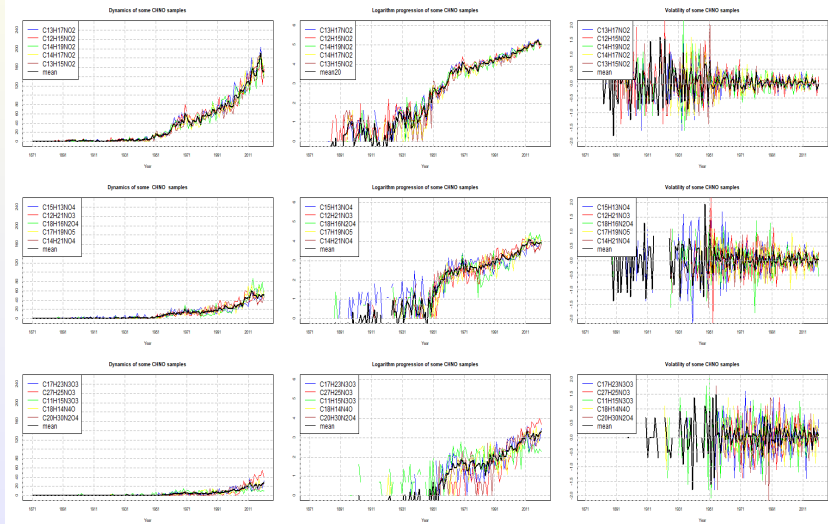
Samples of CH & CHNO



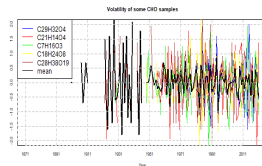
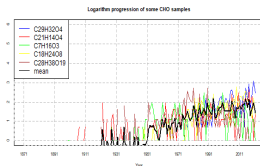
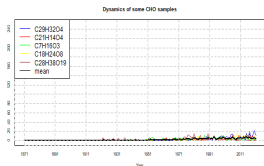
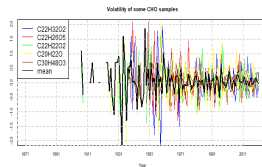
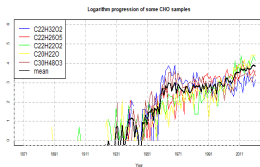
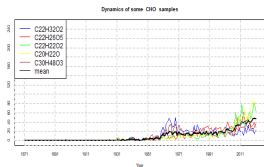
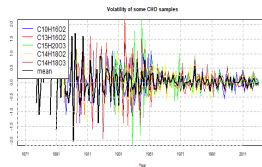
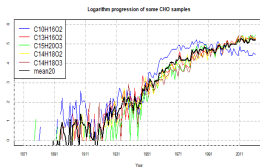
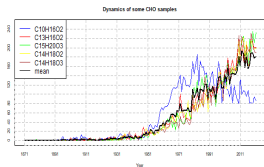
Samples of CHO & CHN



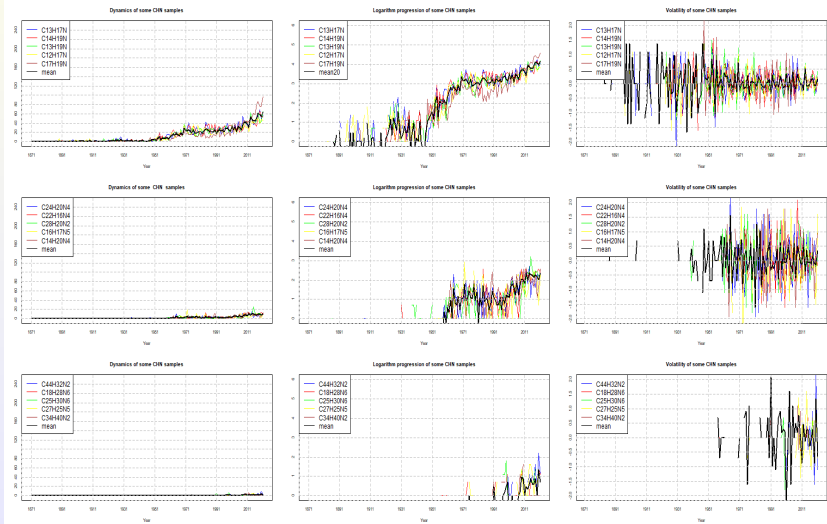
Samples of CHNO formulae



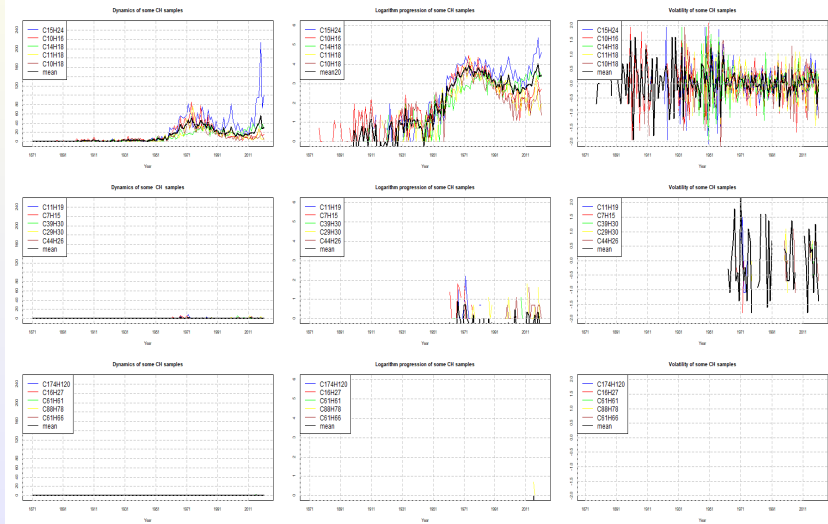
Samples of CHO formulae



Samples of CHN formulae



Samples of CH formulae



Agent-based model

The study of variance analysis for a pool of multiple entities dates back from the work of Markowitz on asset allocation, as a direct consequence of the law of large number. More precisely, let us consider a pool of n entities whose dynamics follow the equation

$$dy_i = \mu_i y_i dt + \sigma_i y_i dB_i(t), i = 1, \dots, n, \quad (0.1)$$

where μ_i, σ_i are corresponding parameters of entity i and B_i is a source of Gaussian noises. Also assume that the weight of entity i in the pool is p_i , i.e. $p_i = \frac{y_i}{V}$ where $V = \sum_{i=1}^n y_i$. Then one can derive the dynamics of V as

$$\frac{dV}{V} = \sum_{i=1}^n \frac{dy_i}{y_i} \frac{y_i}{V} = \left(\sum_{i=1}^n p_i \mu_i \right) dt + \sum_{i=1}^n p_i \sigma_i dB_i(t) \quad (0.2)$$

In case of mutually independent noises B_i and adapted p_i , one can solve the equation (0.2) explicitly so that the logarithmic scale $\log V$ follows the dynamics

$$d \log V = \sum_{i=1}^n \left(p_i \mu_i - \frac{p_i^2 \sigma_i^2}{2} \right) dt + \sum_{i=1}^n p_i \sigma_i dB_i(t). \quad (0.3)$$

Variance analysis

It follows from equation (0.3) and the mutual independence of the noises that the variance of the log-volatility $d \log V$ can be approximated as

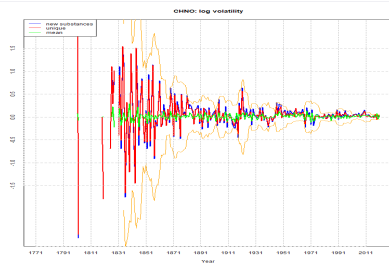
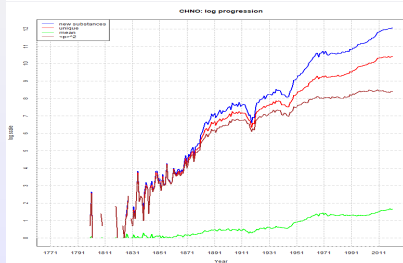
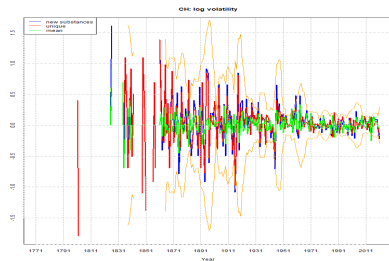
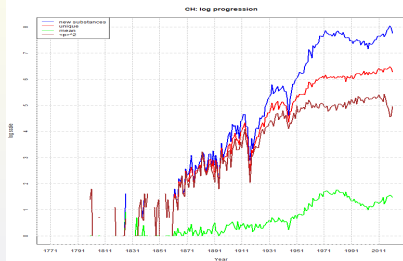
$$\text{Var}(\log V) = \sum_{i=1}^n p_i^2 \sigma_i^2 \text{Var}(dB_i). \quad (0.4)$$

Ideally, if we make an additional assumption on the i.i.d. of B_i and the volatility parameters σ_i are close to a constant, i.e. $\sigma_i \equiv \sigma$ and $\text{Var}(dB_i) \equiv \text{Var}(dB)$, then one deduces from equation (0.4) that

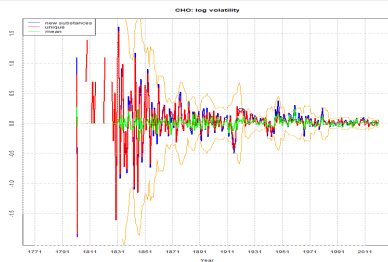
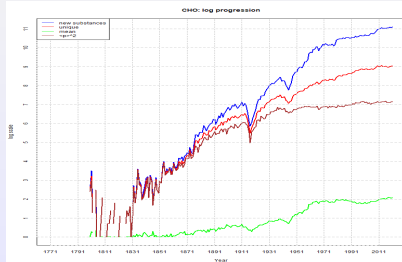
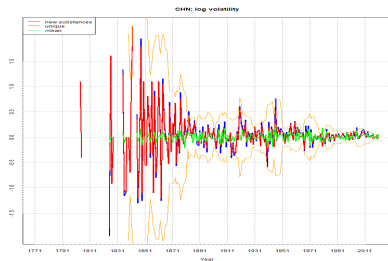
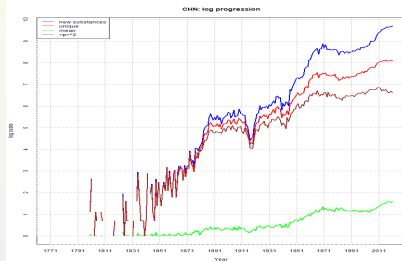
$$\text{Var}(d \log V) = \left(\sum_{i=1}^n p_i^2 \right) \sigma^2 \text{Var}(dB). \quad (0.5)$$

- While it is impractical to assume that p_i stay constant over time, it is ok to assume that (p_i) are mutually independent of (B_i) .
- μ_i, μ should also depend negatively on y , i.e. similar effect to the logistic curve.

Dynamics of CH & CHNO



Dynamics of CHN & CHO



Variance analysis

Since $\sum_{i=1}^n p_i = 1$, it follows that $\sum_{i=1}^n p_i^2 \ll \sum_{i=1}^n p_i = 1$, which implies that the variance of log-volatility of the pool is much less than the variance of the log-volatility of each participating entity. Moreover, equation (0.5) shows that the variance of the pool is going to be reduced as the number n of participating entities grows. Indeed, in the ideal case, for instance, that $\pi_i \equiv p = \frac{1}{n}$, equation (0.5) yields

$$\text{Var}(d \log V) = \frac{1}{n} \sigma^2 \text{Var}(dB), \quad (0.6)$$

which shows the negative effect of the number of entities into the variance of the log-volatility of the pool. In reality, one might not have the i.i.d. of noises B_i , nor the same volatility parameter $\sigma_i \equiv \sigma$, nor the equal weight $p_i \equiv p$, thus the effect (0.6) might be quantitatively different. Yet one should expect the negative effect $\text{Var}(\log V) \approx \frac{1}{n}$ stays the same.

A realistic model has the form: $Y = (y_i)$, $\mathbf{B} = (B_i)$ where

$$dY = AYdt + \Sigma Y \otimes d\mathbf{B}$$

for certain drift and diffusion matrices.

Law of large numbers

In practice, we are dealing with a time dependent model with n_t classes of new substances, each class consists of all new substances of the same formula. As shown above, the dynamics of each class seems to follow the identical distribution $\log y_i \approx_d (\mu, \sigma)$, although it is not sure if the classes are mutually independent. Also, the dynamics of n_t is supposed to be independent of the the dynamics of the classes, and follows the equation

$$d \log n_t = \mu^* dt + \sigma^* d\xi_t \Rightarrow n_t = \exp\{\mu^* t + \sigma^* \xi_t\}.$$

As such $Ed \log V_t = \mu$ The variance of $d \log V_t$ in case of equally allocation then has the form

$$\text{Var}(d \log V_t) = \frac{\sigma^2}{n_t} = \frac{\sigma^2}{\exp\{\mu^* t + \sigma^* \xi_t\}}.$$

This explains the exponential reduction of By the Borel-Cantelli lemma, the log-volatility d_t tends to converge to μ as t tends to infinity.

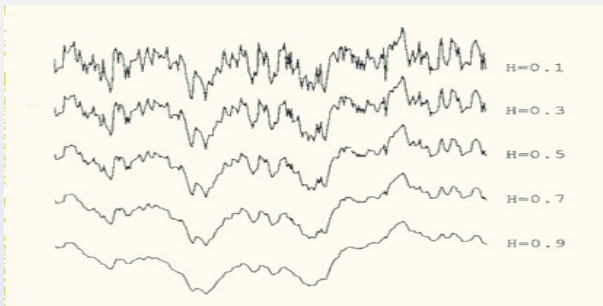
Fractional Brownian motions

- Given $H \in (0, 1)$, fBm is $B^H = (B^H(t), t \in \mathbb{R})$ with covariance function

$$R(s, t) = \frac{1}{2}(|t|^{2H} + |s|^{2H} - |t - s|^{2H}), \quad s, t \in \mathbb{R}.$$

When $H = 1/2$, B is the standard Brownian motion.

- Long range dependence: for $H > 0.5$ $\sum_{k=-\infty}^n \text{cov}(\Delta B^H(n), \Delta B^H(k)) = \infty$.



Long range dependence and similarity

At this step we are going to consider the similarity of distribution over the years. The similarity metric can be computed for two consecutive years and also for each year to the reference year 2021. The formulae of the similarity are as follows.

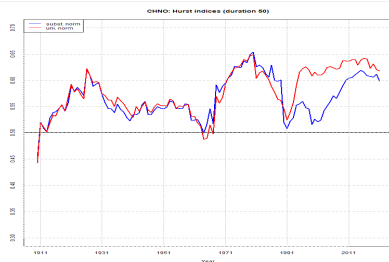
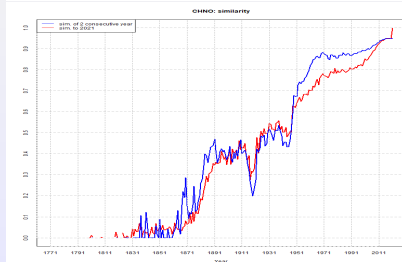
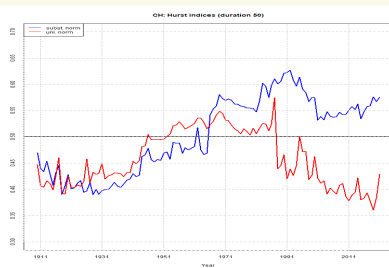
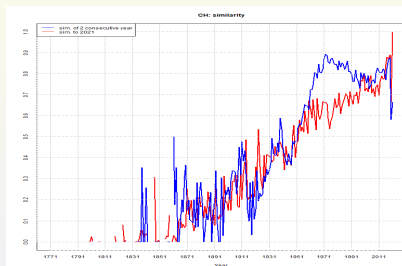
$$Sim_t^1 = \frac{\langle D_t, D_{t+1} \rangle}{\|D_t\|_2 \|D_{t+1}\|_2}$$

and

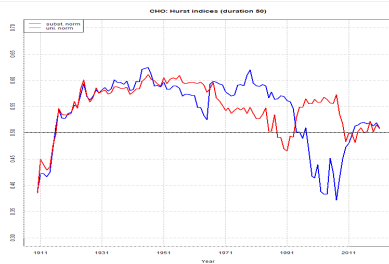
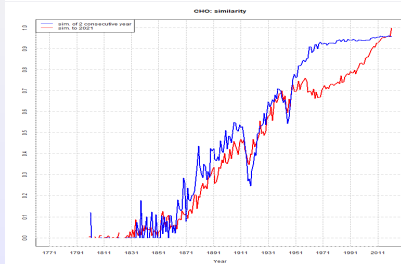
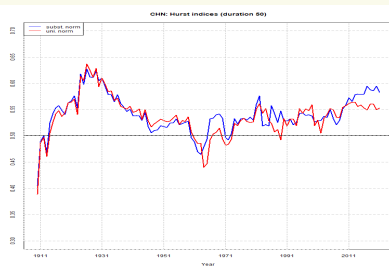
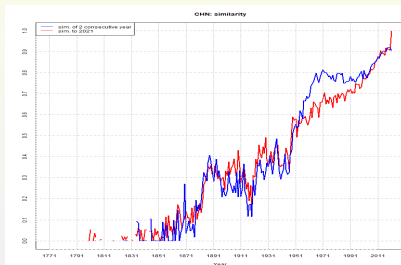
$$Sim_t^2 = \frac{\langle D_t, D_{2021} \rangle}{\|D_t\|_2 \|D_{2021}\|_2}$$

where $\langle \cdot, \cdot \rangle$ and $\|\cdot\|_2$ are respectively the Euclidean inner product and the Euclidean norm.

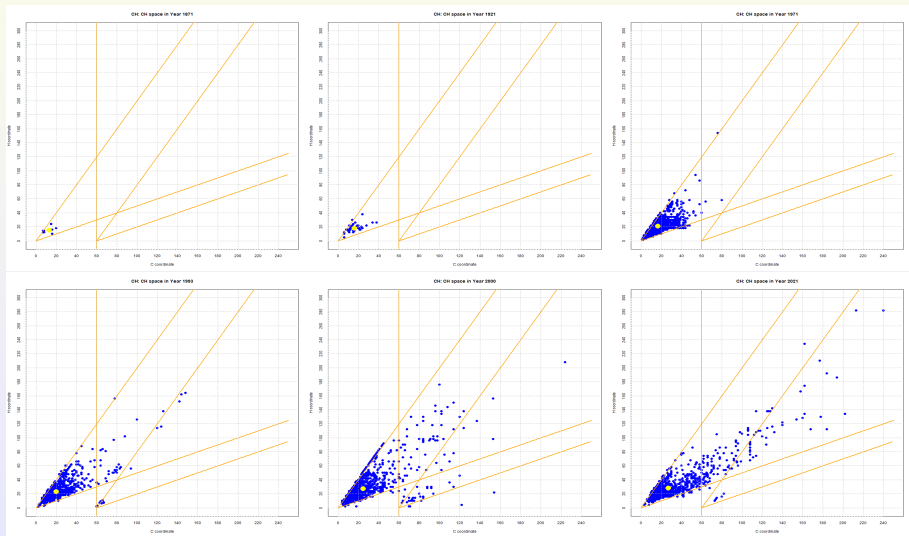
Similarity and Hurst indices of CH & CHNO



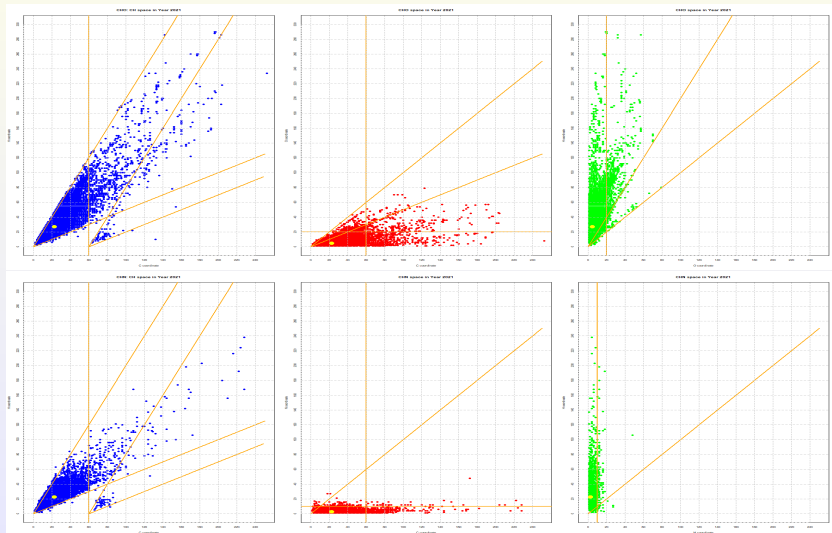
Similarity and Hurst indices of CHN & CHO



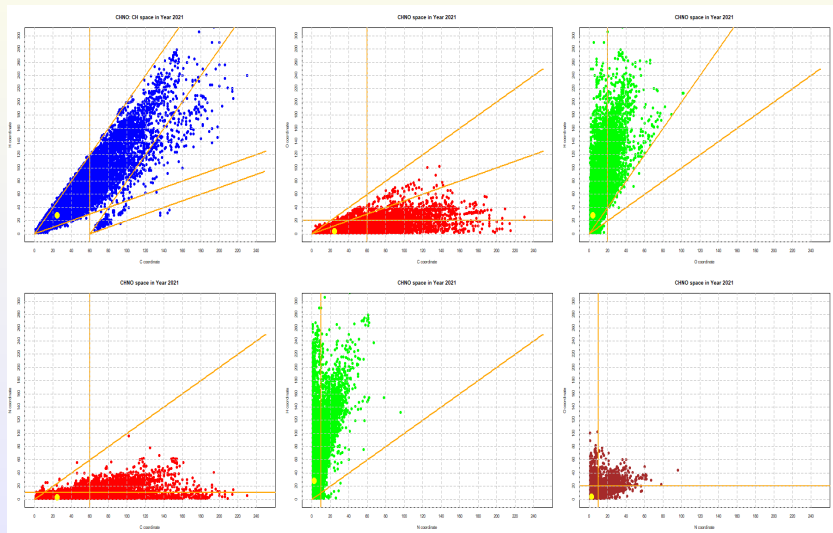
Topology space of CH



Topology spaces of CHO & CHN

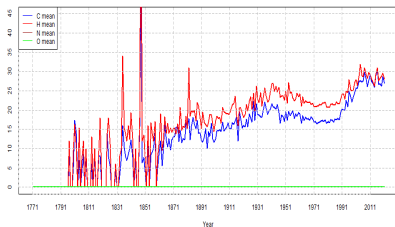


Topology space of CHNO

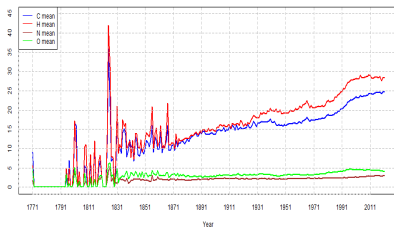


Dynamics of bary-center of CH & CHNO

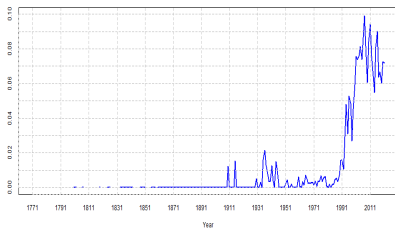
CH: mean progression



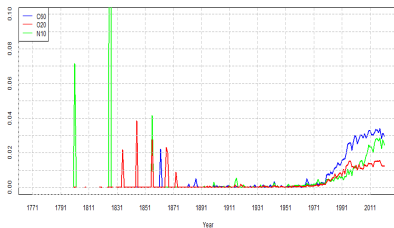
CHNO: mean progression



probability of C thresholds

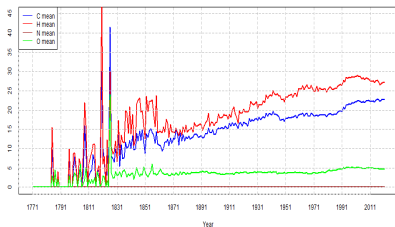


probability of C, N and O thresholds

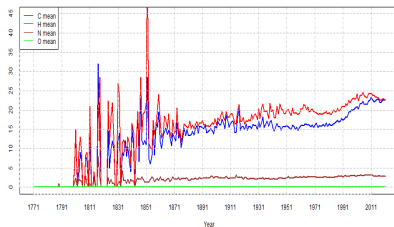


Dynamics of bary-center of CHO & CHN

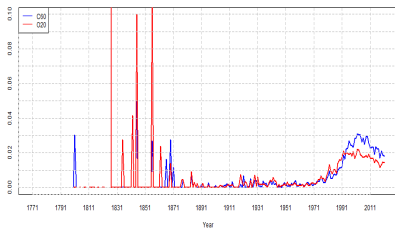
CHO: mean progression



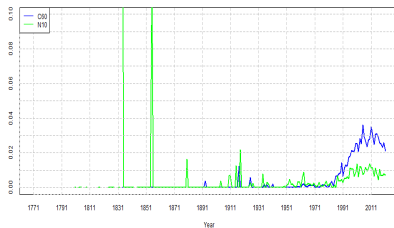
CHN: mean progression



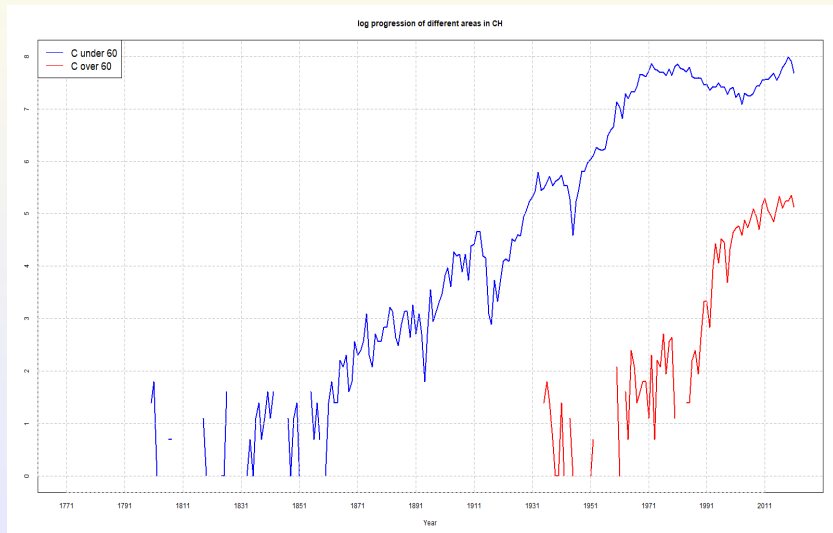
probability of C and O thresholds



probability of C and N thresholds



Dynamics of CH in different areas



Dynamics of bary-center

- While the "C under 60" still dominates the trend in CH, there are innovations since 1975-1980 in the area of "C over 60" with movement toward more complexity, which accounts for 8-10 percent of new substances.
- Meanwhile there is gradual increases in CHN, CHO, CHNO to the new trends, which results in the "C over 60" area to account only a portion of around 3 percent. Similar phenomena happen for the "O over 20" and "N over 10" areas.
- Since 2000, in CH and CHN compounds, one witnesses the C coordinate in the bary-center to approach the H coordinate, indicates that more new substances C_xH_y , $C_xH_yN_z$ with very high $y \approx x$ is found.
- Meanwhile, such phenomena have yet to happen with CHO and CHNO. One might speculate that if the trend of increasing the complexity in CH compounds continues, it finally results in CHO and CHNO compounds by certain combinatoric rules in chemistry.
- Proposal: to call the "C under 60" the **conservative area** and the "C over 60" the **complexity area**.

Distribution via level sets

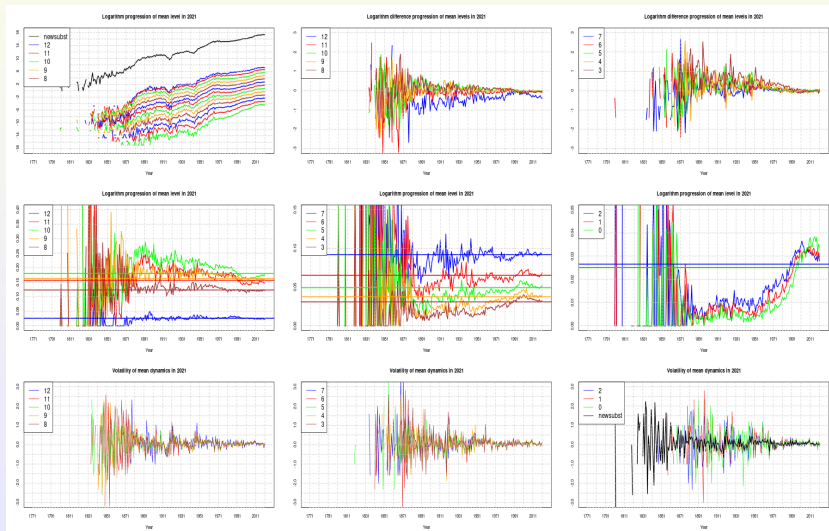
Define $\Omega^k := \{\omega \in \Omega : \log_2(\sum_t \omega_t) \in [k, k+1)\}, k = 1, \dots, 12$.

Observation: Ω_t^k has similar dynamics to Ω_t . As a result, the evolution of the expected value m_t of isomers on level sets Ω^k has the same dynamics as the whole new substances Ω_t .

Explanation: The distribution $p_k = \mathbb{P}(\Omega^k)$ looks stationary. Because: if $\frac{a}{b} = \frac{c}{d}$ then

$\frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$. More generally, $\frac{\Omega_t^k}{\Omega_t} \approx \frac{\sum_t \Omega_t^k}{\sum_t \Omega_t} = p_k$.

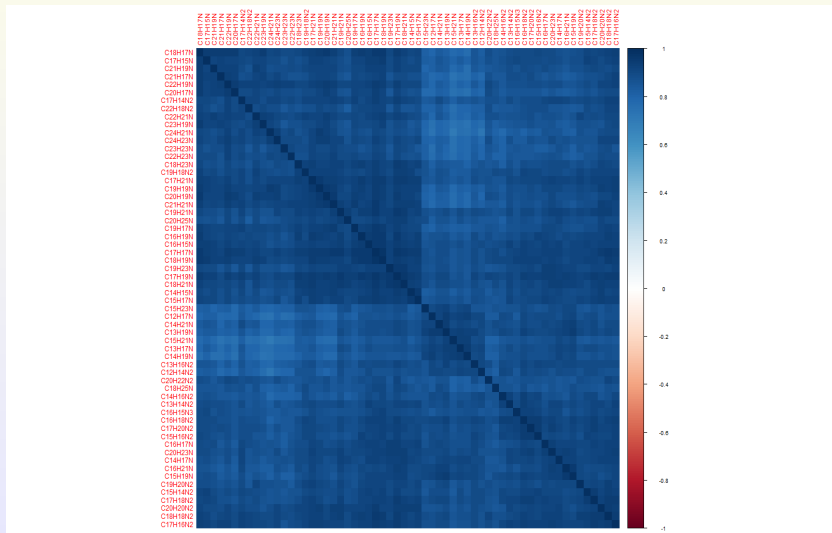
Distribution via level sets



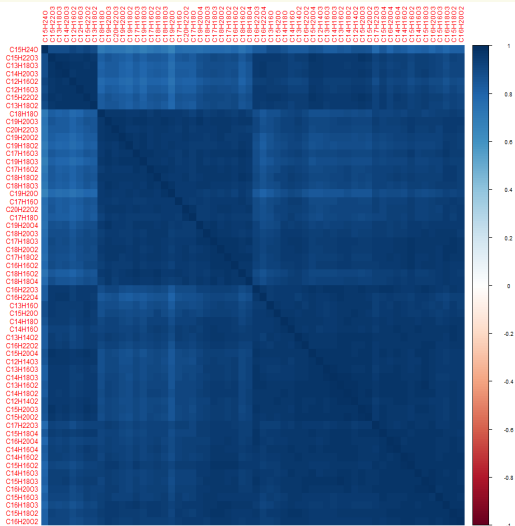
Thank you for your attention!



Correlation matrix of CHN



Correlation matrix of CHO



Correlation matrix of CHNO

